



Insights to antimicrobial resistance: heavy metals can inhibit antibiotic resistance in bacteria isolated from wastewater

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Abstract The alarming upsurge in the co-existence of heavy metal and antibiotic resistance may have a devastating impact on humans, animals, and the environment. Four metal-resistant bacteria were isolated from hospital effluents and industrial drain. Heavy metal resistance and antimicrobial resistance were examined in the isolates followed by identification through 16S rRNA gene sequencing. *Delftia tsuruhatensis* strain FK-01 and *Carnobacterium inhibens* strain FK-02 tolerated arsenic with maximal tolerated concentration (MTC) of 30 mM and 10 mM, respectively. *Staphylococcus hominis* strain FK-04 tolerated copper up to 4 mM and lead-resistant *Raoultella ornithinolytica* strain FK-05 exhibited tolerance to 1 mM lead. The growth kinetics of bacteria were monitored in the presence of metals and the following antibiotics, tetracycline, chloramphenicol, and kanamycin. The presence of arsenate significantly enhanced tetracycline resistance in *C. inhibens*. Heavy metal-induced antibiotic resistance was also observed in *S. hominis* and *R. ornithinolytica*, against chloramphenicol and tetracycline respectively. *D. tsuruhatensis* showed resistance to kanamycin but when grown in the presence of arsenic and kanamycin, bacteria lost resistance to the antibiotic. Therefore,

it is suggested that the novel arsenate-resistant strain *Delftia tsuruhatensis* FK-01 has a unique ability to inhibit antimicrobial resistance that can be harnessed in bioremediation.

Keywords Heavy metal-resistant bacteria · Antibiotic resistance · Arsenic · Lead · Copper · Water

Introduction

Heavy metal pollution has become a major worldwide issue because of toxicity to humans, animals, and the environment. Water contamination from heavy metals represents a considerable threat to global health. A major driver of water pollution is the industrial wastewater (Mokarram et al., 2020). Many sources can significantly contribute towards metal contamination in the environment such as agriculture manure, industrial effluents of electroplating, petrochemical, paint, leather, and tanning industries (Tokar et al., 2011). Heavy metals enter the ecosystem as persistent and non-degradable contaminants (Lu et al., 2020). Without proper wastewater filtration, the release of heavy metals (HMs) from the effluent waste can cause immense health problems due to their persistence, bio-magnifications, and accumulation in the food chain (Singh & Kalamdhad, 2011).

Antimicrobial resistance has become a global issue. Multidrug-resistant (MDR) strains are causing deaths

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of thousands of people each year (Tóth et al., 2020). The inappropriate use of antibiotics in daily life is also releasing them in the environment and promoting subsequent development of antibiotic resistance in microorganisms (Segura et al., 2009; Khan et al., 2013). Factors that can significantly contribute to the burden of antibiotic resistance include hospital waste, wastewater treatment plants, sewage treatment plants, pharmaceutical wastage, the misuse of antibiotics in the veterinary industry, agriculture, animal husbandry, and aquaculture (Martinez, 2009; Nikolaou et al., 2007). Antimicrobial resistance genes (ARGs), found in a variety of samples, are responsible for the development of antimicrobial resistance. Increasing evidence suggests that antimicrobial resistance genes (ARGs) carried by non-pathogenic bacteria can be transferred to pathogens by horizontal gene transfer (HGT) further promoting resistance against antibiotics (Tóth et al., 2020). The problem of developing antibiotic resistance has become a more frequent topic of research groups and their studies around the world (Li & Webster, 2018). This issue is very well researched at the level of antimicrobial resistance but has little information on combined effect of AMR and heavy metals (Lu et al., 2020). HGT is involved in the selection of bacterial strains which are resistant to the metal polluted environment and capable of bioremediation (Biswas et al., 2021). Besides, the involvement of heavy metals in the proliferation of antibiotic resistance genes (ARGs) by co-selection and cross-resistance has also been reported (Baker-Austin et al., 2006). In water, bacteria from various sources, such as human, animal, and environment, can merge and transfer resistance genes via HGT, mobile genetic elements (MGEs), and vectors that further spread resistance (Gheraout & Elboughdiri, 2020). It should be noted that the combined use of different metal salts and antimicrobials plays an important role in the transfer of resistance genes in microorganisms and may contribute to the development of multidrug-resistant superbugs. For instance, zinc ion co-exposure with the antibiotic oxytetracycline in industrial effluents resulted in increased antibiotic resistance (Peltier et al., 2010). Both nickel and cadmium metals reportedly enhanced resistance in freshwater microcosms to chemically unrelated antibiotics like ampicillin and chloramphenicol (Stepanuskas et al., 2006). A recent study of Changshou Lake showed a significant correlation between antibiotics and HMs, manganese, and zinc (Lu et al., 2020). In agricultural soil, the addition

of copper was responsible for copper resistance and also co-selection for tolerance towards multiple antibiotics such as ampicillin, chloramphenicol, and tetracycline (Berg et al., 2005). Recently, use of HMs in poultry farms for promoting growth caused co-selection for MGEs and ARGs (Mazhar et al., 2021). Substantial heavy metals are persistent in our ecosystem; therefore, the indigenous microbiota gets exposed to several types of metals such as arsenic, lead, cadmium as well as antibiotics at the same time. Gut microbiota participates in several physiological functions with profound effects on health. Environmental pollutants including HMs and antibiotics cause alteration in the composition of microbiome which has been associated with disorders of energy metabolism, nutrient absorption, and immune system (Jin et al., 2017). Additionally, the mixed pollutants in the ecosystem are creating an alarming situation, regardless of the combined impact concerning the selective ability of antibiotic resistance (Skurnik et al., 2010). One of the predictable reasons of the rise in antibiotic resistance may be the high concentration of metals in industrial discharge which can act as a common stress element and could be responsible for the emergence of new antibiotic resistance determinants when both of them interact together (Martinez & Rojo, 2011).

In the present study, four metal-resistant bacteria, *Delftia tsuruhatensis* strain FK-01, *Carnobacterium inhibens* strain FK-02, *Staphylococcus hominis* strain FK-04, and *Raoultella ornithinolytica* strain FK-05 were isolated from wastewater samples (Table 1). The co-existence between heavy metal resistance (HMR) and antimicrobial resistance (AMR) was observed in the isolates.

Results

Identification of bacteria

In this study, four new metal-resistant strains were isolated (Table 2, Fig. 1). Two arsenic-resistant bacteria, *D. tsuruhatensis* strain FK-01 and *C. inhibens* strain FK-02, one copper resistant *S. hominis* strain FK-04, and one lead resistant *R. ornithinolytica* strain FK-05 were identified (Fig. 2). GenBank accession numbers obtained for the 16S rRNA gene sequences of each strain are listed in Table 2.

Table 1 Chemical analysis of wastewater samples

Properties	Industrial samples (mg/L)	Hospital samples (mg/L)
Potassium	13.97	11.55
NO ₃	106.69	114.41
PO ₄	1.38	1.11
SO ₄	42.54	54.88
Calcium	68.5	216
Sodium	88.22	75.25
Magnesium	42.2	49.2
pH	8.34	8.32

Maximal tolerated concentration of isolates

To find out maximal tolerated concentration (MTC), each isolate was given a different concentration of metal pressure by which 90% bacterial growth stopped. Figures 3 and 4 represent the MTC of isolates. It is worth mentioning, when the concentration of each metal increases, the bacterial growth decreased accordingly. Briefly, the response of each isolate is given under.

D. tsuruhatensis grew well at 2 mM to 16 mM metal concentrations as compared to control. With the exposure to high concentration of 32 mM arsenate, the lag phase extended from 0 to 48 h. Bacteria entered exponential phase after 48 h which is significantly better (Fig. 3a). *C. inhibens* strain FK-02 was able to grow at arsenate concentrations of 2 mM to 8 mM. However, growth was retarded with 12 mM concentration. MTC was observed as 10 mM (Fig. 3b).

S. hominis strain FK-04 was isolated as copper resistant in this study. Bacteria entered exponential phase within 3 h with 0.5 mM to 2 mM metal concentrations as compared to control. No growth was observed with 12 mM copper after 56 h. MTC was determined as 4 mM (Fig. 4a). Lastly, lead-resistant *R. ornithinolytic* strain FK-05 showed MTC of 1 mM. Bacteria were unable to grow at higher concentrations of 2 mM and 4 mM after 48 h of incubation (Fig. 4b).

Investigation of the co-existence of heavy metal resistance and antimicrobial resistance

The metal-resistant isolates were grown in the presence of antibiotics to study the effects of heavy metal resistance (HMR) over antimicrobial resistance (AMR). The behavior of each organism in the presence of antibiotics plus metal was quite different when the same organism was only grown with metals. The response of each strain is given under.

Delftia tsuruhatensis strain FK-01

The growth curve shows the response of *D. tsuruhatensis* in the presence of arsenic and antibiotics, Tet, CMP, and Kan (Fig. 5a). The bacteria were not resistant to Tet and CMP but displayed natural resistance to Kan and grew well initially with a decline after 24 h. No antimicrobial resistance was induced due to the presence of As. Remarkably, the presence of arsenic inhibited Kan resistance so that no growth could be observed with As and Kan both in the culture (Fig. 5a). This is the first report to describe inhibition of AMR due to As.

Table 2 Morphological characterization of bacteria isolated from hospital and industrial discharge

Isolate ID	Wastewater source	Colony	Gram	Morphology
<i>Delftia tsuruhatensis</i> strain FK-01 (GenBank accession number MW032203)	Hospital discharge	White	Negative	Rod
<i>Carnobacterium inhibens</i> strain FK-02 (GenBank accession number MW032204)	Industrial discharge	Creamy mucoid	Positive	Rod
<i>Staphylococcus hominis</i> strain FK-04 (GenBank accession number MW032206)	Hospital discharge	White	Positive	Cocci
<i>Raoultella ornithinolytica</i> strain FK-05 (GenBank accession number MW032207)	Industrial discharge	Creamy mucoid	Negative	Rod

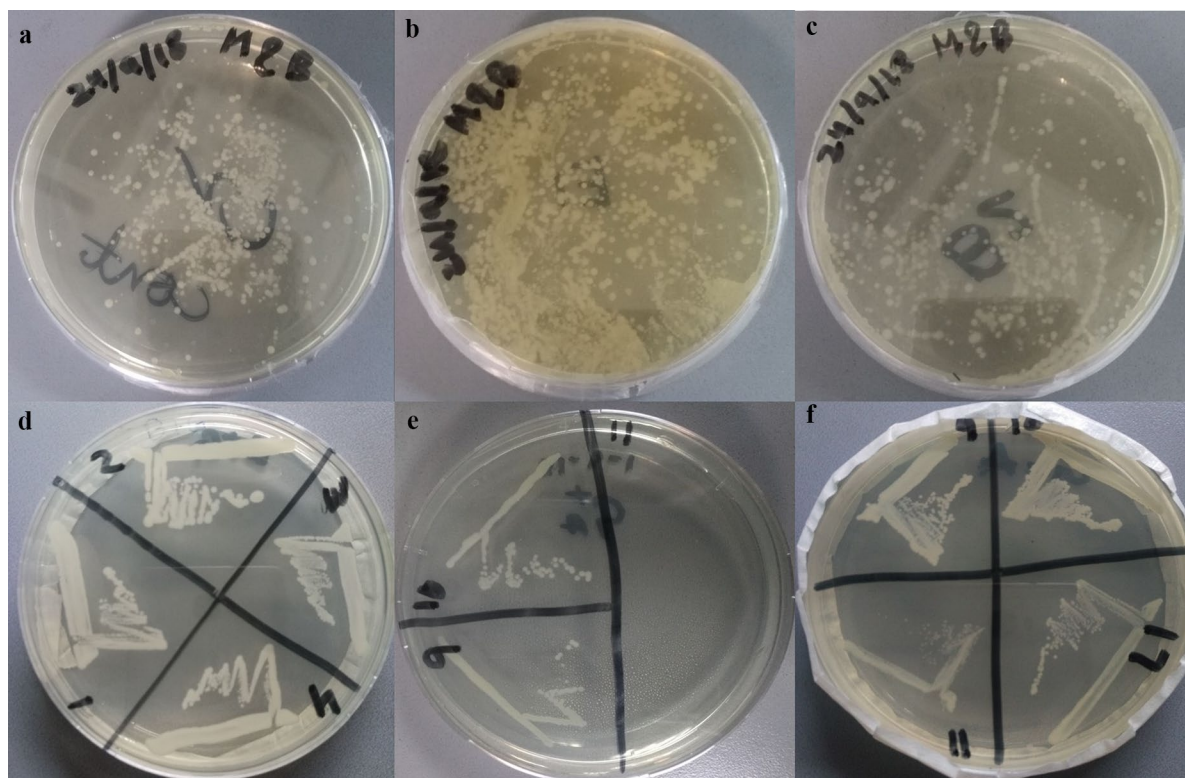


Fig. 1 Heavy metal-resistant isolates were cultured on nutrient agar (NB) plates. **a, b, c** A few culture plates are shown. **d, e, f** Isolates were streaked to obtain pure cultures

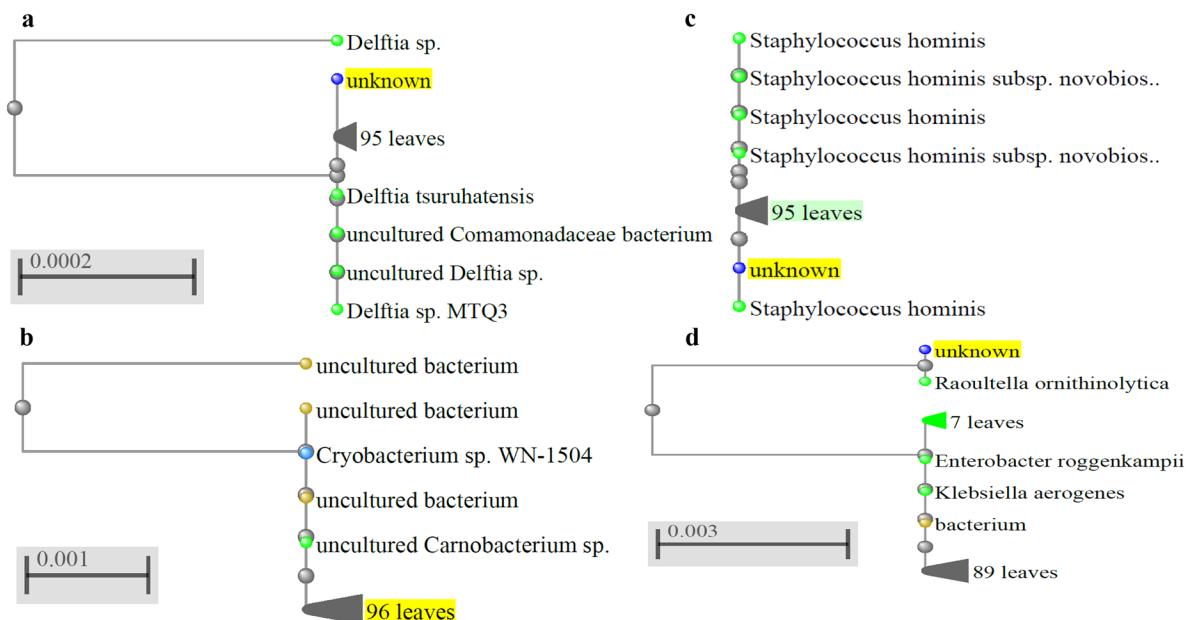


Fig. 2 Phylogenetic tree showing the homologs of strains. **a** *Delftia tsuruhatensis* strain FK-01, **b** *Carnobacterium inhibens* strain FK-02, **c** *Staphylococcus hominis* strain FK-04, **d** *Raoultella ornithinolytica* strain FK-05

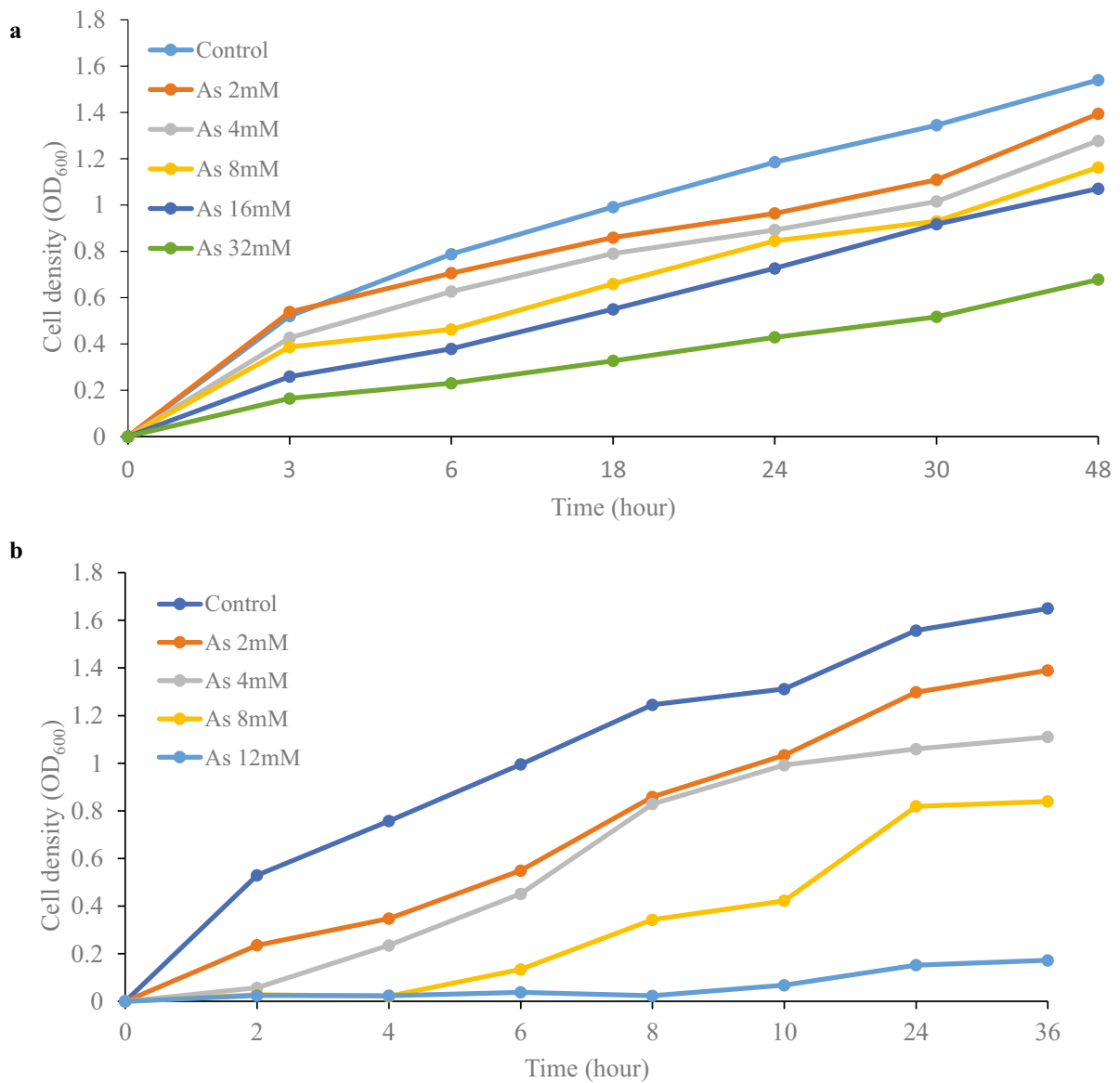


Fig. 3 MTC of arsenic-resistant isolates. **a** The response of *Delftia tsuruhatensis* strain FK-01 in the presence of arsenic. **b** *Carnobacterium inhibens* strain FK-02 in the presence of arsenic

Carnobacterium inhibens strain FK-02

Figure 5b represents the response of *C. inhibens* with antibiotics, Tet, Kan, and CMP in the presence of arsenic. The growth curve shows that the bacteria were not resistant to any of the antibiotics and did not grow (Fig. 5b). It was observed that the presence of arsenic in the growth media induced tetracycline resistance after 24 h of incubation and bacteria were able to grow successfully up to 144 h, until the end of the experiment.

Staphylococcus hominis strain FK-04

Figure 6a shows the growth profile of *S. hominis* in the presence of copper and antibiotics (Tet, CMP, and Kan). The bacteria were resistant to all antibiotics and could grow well except that the presence of Kan caused slow growth. The induction in antibiotic resistance was much more pronounced when Cu was supplemented with CMP and Tet. The resistance to CMP was induced with a huge boost in the growth with CMP and Cu which was not observed in the case

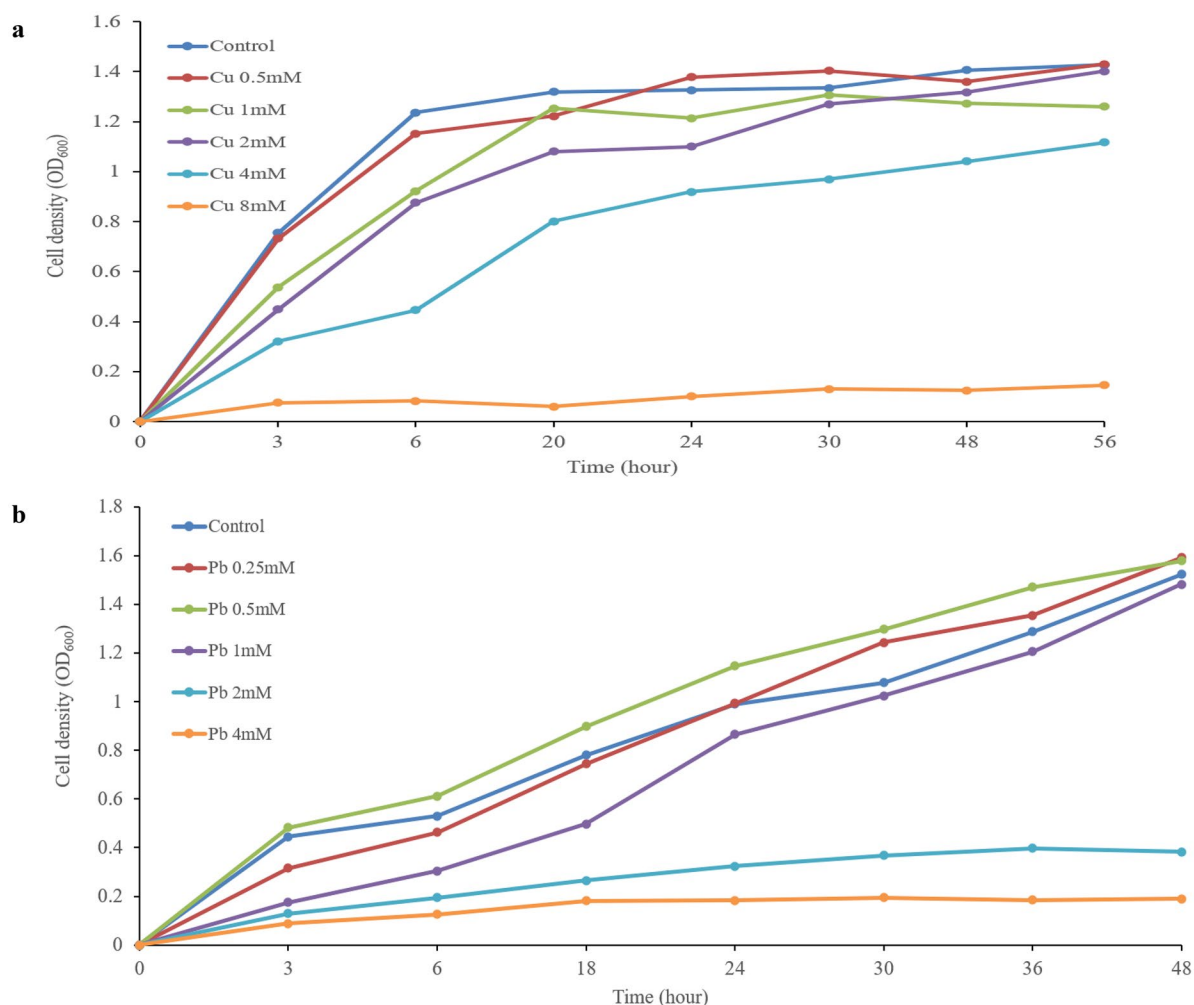


Fig. 4 MTC of copper and lead resistant isolates. **a** MTC of *Staphylococcus hominis* strain FK-04 in the presence of copper. **b** MTC of *Raoultella ornithinolytica* strain FK-05 in the presence of lead

of CMP alone. It was also seen that the bacteria were highly resistant to Tet and the presence of Cu induced even more resistance to Tet (Fig. 6a).

Raoultella ornithinolytica strain FK-05

The growth kinetics of *R. ornithinolytica* were observed in the presence of lead and antibiotics (Fig. 6b). Bacteria were not resistant to Tet and CMP as they were unable to grow in the presence of these antibiotics. When grown with antibiotics and lead, bacteria acquired resistance to Tet. Additionally, bacteria had inherent resistance to Kan and grew well with Kan and lead (Fig. 6b).

Discussion

In this study, four metal-tolerant bacteria were isolated from wastewater samples. *D. tsuruhatensis* strain FK-01 was isolated from hospital wastewater in the present study. *D. tsuruhatensis* was first isolated in 2003 from domestic wastewater in Japan (Shigematsu et al., 2003). The novel strain FK-01 was highly resistant to As tolerating up to 32 mM and had a potential to grow at even higher metalloid concentration. Although, bacteria had resistance to kanamycin, interestingly, when grown in the presence of arsenic and kanamycin, *D. tsuruhatensis* tolerated such a high concentration of arsenic and inhibited the kanamycin resistance. This is a new promising finding as

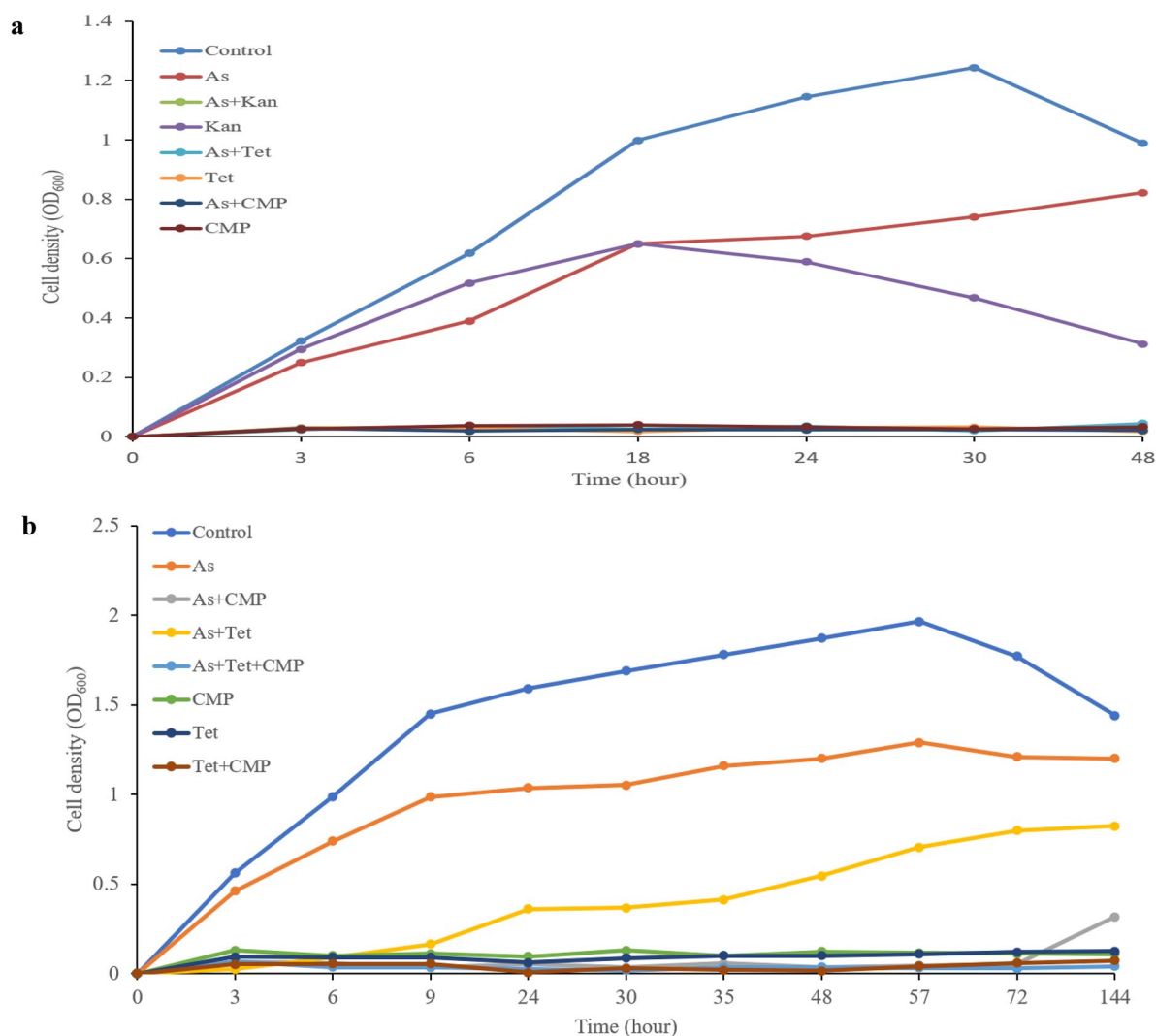


Fig. 5 Growth kinetics of arsenic-resistant bacteria in the presence of antibiotics. **a** Growth curve of *Delftia tsuruhatensis* strain FK-01 in the presence of 2 mM arsenic and Tet,

CMP, Kan. **b** Growth curve of *Carnobacterium inhibens* strain FK-02 in the presence of 2 mM arsenic and antibiotics, Tet, CMP, Kan

these capabilities were not reported earlier. For bioremediation, it is pertinent to use non-pathogenic bacteria which are not resistant to antibiotics. Therefore, it is suggested that the arsenate resistance of *D. tsuruhatensis* and inhibition of the antibiotic resistance prove it a novel tool for bioremediation. We aim to determine the potential abilities of *D. tsuruhatensis* strain FK-01 in future studies. *D. tsuruhatensis* strains with different capabilities have been identified recently. For instance, a denitrifying strain ZK-1 was isolated from activated sludge (Li et al., 2020). Another *Delftia* strain CRS1243 was identified in the

clinical specimen producing IMP-1 (imipenemase 1) metallo- β -lactamase (MBL) (Cho et al., 2021).

The second isolate was *C. inhibens*, a gram-positive facultative anaerobe, lactic acid bacteria previously isolated from sources such as intestine of Atlantic salmon, marine sponges, marine sediments, permafrost, Antarctica, raw foods, dairy products, and quite recently from the blood culture of immunocompromised patient (Leisner et al., 2007; Lo & Sheth, 2021; Nicholson et al., 2013; Romaniuk et al., 2018). *Carnobacterium* spp. have been approved by US Food and Drug Administration (FDA) and Health Canada as a

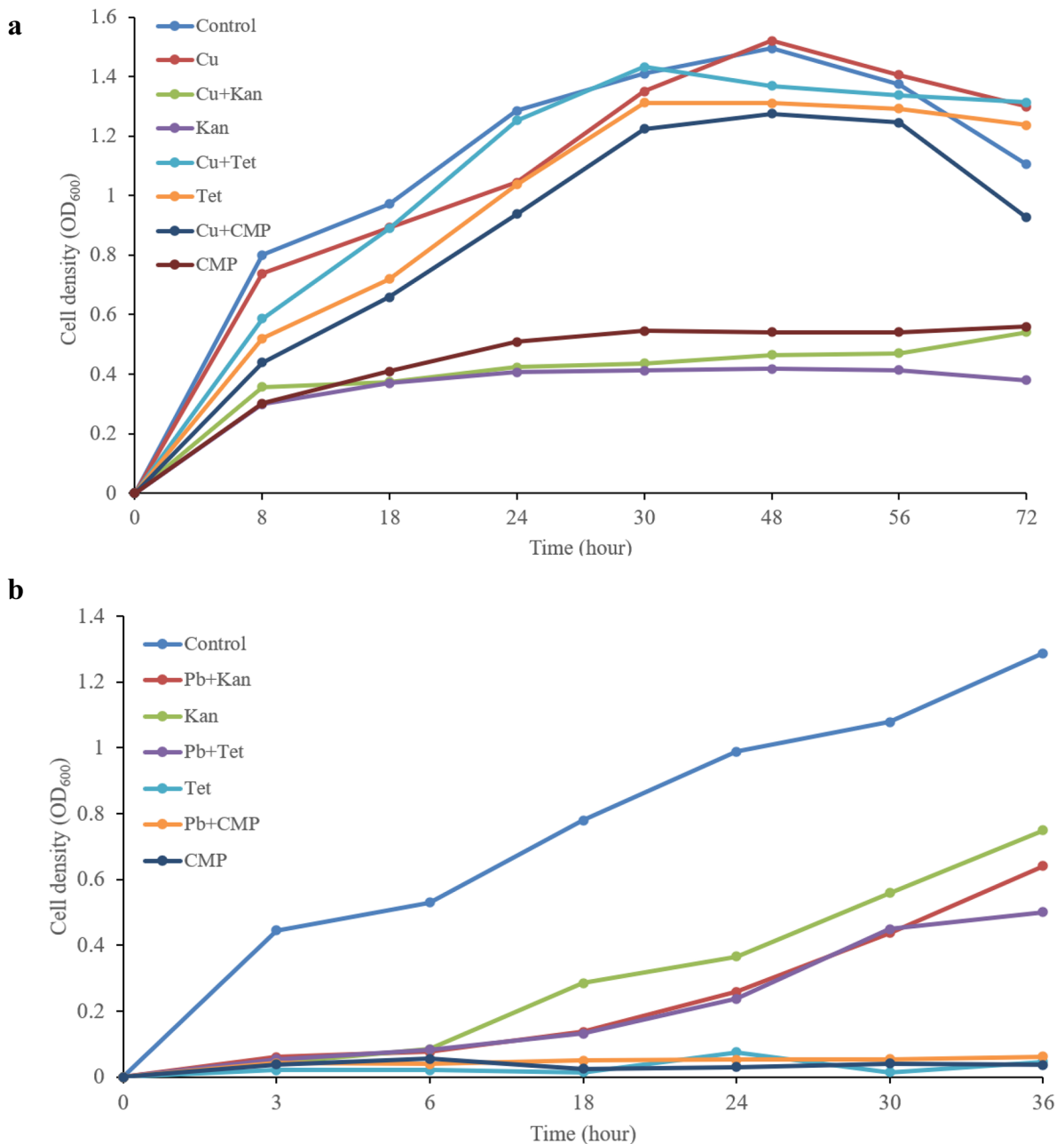


Fig. 6 Growth kinetics of copper and lead resistant bacteria in the presence of antibiotics. **a** Growth curve of *Staphylococcus hominis* strain FK-04 in the presence of 0.5 mM copper and

Tet, CMP, Kan. **b** Growth curve of *Raoultella ornithinolytica* strain FK-05 in the presence of 0.5 mM lead and Tet, CMP, Kan

bio-preservative in food (Lo & Sheth, 2021). The isolation of *C. inhibens* strain from industrial wastewater is reported for the first time in the present study, such a source has never been reported earlier. *Carnobacterium* tolerated As up to 8 mM significantly but when the concentration was increased up to 12 mM the

growth of bacteria decreased up to approximately 90% (MTC90). Although, heavy metals resistance was previously observed for *Carnobacterium* but MTC was not worked out in detail (De La Rosa-Acosta et al., 2016). Regarding the antimicrobial resistance, *Carnobacterium* was not known to possess resistance to

tetracycline. For the first time, the organism was tested against antibiotics in the presence of arsenic. Interestingly, cross-resistance occurred, the growth of *C. inhibens* strain FK-02 was significantly promoted in the presence of As (2 mM) and tetracycline (12.5 µg/mL) as compared to the tetracycline alone indicating the induction of tetracycline resistance (Fig. 4a). This co-existence mechanism involving arsenic and tetracycline resistance has not been reported yet, which is a new finding for *C. inhibens*. This organism has the capacity to inhibit the growth of the fish pathogens like *Vibrio anguillarum* and *Aeromonas salmonicida* (Ringø, 2004). Furthermore, the organic arsenicals have been used as growth promoters in fish farming; therefore, it is suggested that this organism may be useful in fish farms as a probiotic due to high As tolerance and the production of lactic acid could be helpful for fishes to better feed and survive against harsh environment in fish farming (Kristiansen et al., 2011; Ringø, 2008).

Both studied bacteria were highly resistant to arsenic which could be due to high frequency of plasmids in arsenic resistance bacteria as compared to natural bacteria or could be due to the presence of arsenic resistance genes in bacterial chromosomes such as *arsRDABC* and *arsRBC* (Cheng et al., 2021; Lin et al., 2007; Zolgharnein et al., 2007), while the exact resistance mechanism is still unclear.

S. hominins strain FK-04, coagulase-negative, gram-positive bacteria belonging to family *Staphylococcaceae* was isolated as copper resistant organism. Previously, it was considered as a harmless commensal on animal and human skin. However, many reports revealed emerging multidrug resistance that can cause severe septicemia and urinary tract infection (UTI) in cancer patients and other skin infections (Ahmed et al., 2017; Qiao et al., 2013). In this study, bacteria tolerated Cu^{+2} up to 4 mM (Fig. 3a) which is not reported yet. Similar results have been reported in different studies in which coagulase-negative *Staphylococcus* species causing infections in humans showed resistance against different heavy metals such as Pb, Cd, and Cr (Lawal et al., 2021; Ugur & Ceylan, 2003). Furthermore, chloramphenicol was the only antibiotic which showed cross-resistance process, as the chloramphenicol resistance got enhanced in the presence of Cu and this property is not described in any other studies (Fig. 5). Bacteria were already resistant to the other two antibiotics tetracycline and

kanamycin, even without metal addition. This demonstrates multidrug resistance (MDR) property of *S. hominins* which was also reported previously (Ahmed et al., 2017; Palazzo et al., 2008; Roy et al., 2014).

The last isolate of our study, *R. ornithinolytica* strain FK-05, gram-negative bacillus belongs to family *Enterobacteriaceae* was identified as lead tolerant. This organism can cause different kinds of infections in humans such as UTI, gastrointestinal infections, wound and skin infections, bacteremia, pneumonia, osteomyelitis, meningitis, cerebral abscess, mediastinitis, and sepsis (Seng et al., 2016). Recently, *R. ornithinolytica* strains were isolated from blood culture in calf (Pas et al., 2021), chicken products (El-Shannat et al., 2020), and retail ground meat products (Ballash et al., 2021). In this study, strain FK-05 tolerated up to 2 mM lead and at a higher concentration of 4 mM, growth suddenly stopped due to high toxicity of lead. *R. ornithinolytica* was previously reported to tolerate eleven different types of metals including lead, copper, nickel, and silver and also regarded as MDR, with resistance to almost every antibiotic tested (Hajjar et al., 2020; Koc et al., 2013; Mal et al., 2019). In the present study, the cross-resistance in *R. ornithinolytica* was identified in the presence of 0.5 mM copper with tetracycline. The bacterial growth was significantly promoted in the presence of lead with tetracycline when compared to the growth with the same quantity of tetracycline alone. This resulted in increased resistance to tetracycline in the presence of lead. This study explored the cross-resistance mechanism for *R. ornithinolytica* for the first time. Furthermore, *Raoultella* showed high resistance to kanamycin without any metal incorporation while chloramphenicol was found to be sensitive.

The present study showed that it is necessary to study bacterial cross-resistance in detail, as the metal and antibiotics are usually used together in poultry, fish, and swine farms. The gut microbiome is exposed to metals and antibiotics simultaneously, exerting selection pressure leading to the resistance towards antibiotics subsequently causing infections (Webber & Piddock, 2003). Moreover, copper, lead, and zinc are also used as a fertilizer in agriculture and widely applied in leather industries, constructions, healthcare, and other areas from where they are released with antibiotics into the environment (Wei & Yang, 2010). The combined use of antibiotics and heavy metals could enhance antibiotic resistance in bacteria

and could further emerge as a global challenge of AMR.

Conclusion

The bacteria isolated from wastewater were novel strains resistant to the high concentration of heavy metals used in this study. Heavy metal-induced AMR was observed in the isolates except arsenic-resistant *Delftia tsuruhatensis* strain FK-01. According to World Health Organization (WHO), AMR is one of the top ten threats to humanity. Therefore, metal-induced AMR restricts the use of bacteria for bioremediation. This study showed for the first time that *D. tsuruhatensis* strain FK-01 inhibited AMR in the presence of arsenic, making it a promising tool for bioremediation studies. It is also proposed that the application of *D. tsuruhatensis* in fish farms can prove beneficial wherein organic arsenicals are used for growth promotion as the arsenic resistance capability of bacteria could obstruct deleterious effects on fishes. The current study emphasizes the need to study the heavy metals induced AMR as it poses great risks to public health. The resistance genes conferring metal resistance and antibiotic resistance should be identified to explore the mechanisms. Protective measures are recommended in combination therapies involving metals and antibiotics, as such emerging resistance could have problems in treating infectious diseases. Further research investigating the co-existence of AMR and HMR is warranted.

Materials and methods

For bacterial isolation, wastewater samples were collected from industrial (33° 39' 24.2" N 73° 03' 22.4" E) and hospital (33° 42' 14.7" N 73° 03' 18.4" E) discharge in Islamabad, Pakistan. Sampling was conducted from industrial sites releasing organic industrial wastewater. Hospital discharges were recognized as hybrid wastewater carrying domestic, industrial, and medical effluents (Emmanuel et al., 2002). From each drain, 3 samples were picked up in sterilized bottles and mixed to form a single heterogeneous mixture and chemical properties were observed (Table 1). For each location, the distance between sites was kept at least 1000 m apart (Khan et al., 2013). Finally, the

serial dilutions of samples were prepared up to 10^{-5} in distilled water to make total 5 mL, 0.1 mL was spread on culture plates (Lucious et al., 2013).

Isolation of metal(loid) resistant colonies

Metal salts, copper sulfate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Cu^{+2}), lead nitrate $\text{Pb}(\text{NO}_3)_2$ (Pb^{+2}), and sodium arsenate $\text{Na}_2\text{HAsO}_4 \cdot 7\text{H}_2\text{O}$ (As^{+5}) were purchased from Sigma-Aldrich. The stock solutions were prepared in double distilled water, filter sterilized using a 0.22 μm syringe filter and stored at 4 °C. Various metal concentrations were supplemented in nutrient agar media to isolate highly resistant organisms and plates were incubated at 37 °C for 24–48 h (Samanta et al., 2012; Schmidt & Schlegel, 1994). At minimum metal concentrations (0.5 mM), numerous bacterial colonies grew on the plate, but when the concentration of respective metals was increased, the number of colonies started to reduce. The concentration was increased until 10 to 15 colonies remained on the plate among which morphologically different colonies were selected for further analysis. The maximum concentration was established for each metal after which further addition did not allow any growth. Metal concentrations of 0.5 mM to 36 mM were used to isolate arsenic-resistant bacteria. For lead and copper resistant strains, 0.5 mM to 4 mM metal concentrations were used. The selected colonies were sub-cultured using the same media for purification. The control experiments were also run in the same manner, but without metal addition to check the viability of the samples.

Metal(loid) resistance of isolates

Growth dynamics of metal-resistant bacteria were monitored by liquid culture assay (Chen et al., 2015) and maximal tolerated concentration (MTC) was calculated for each metal. Two-fold serial dilutions of the metal solutions were prepared in NB broth to establish a concentration gradient comprising of minimum to maximum concentrations. Cultures were incubated at 37 °C. At fixed time intervals, the optical density (OD_{600}) of bacterial cells was measured through spectrophotometer (Agilent Technologies 55AA). Each treatment was comprised of three replicates and control experiments were also set up in which no metals, but only bacterial cells were added.

Investigation of the antibiotic resistance in presence of copper, lead, and arsenic

To find out, whether the antibiotic resistance in bacteria is a function of the presence of heavy metals, the growth kinetics were drawn in the presence and absence of the antibiotics tetracycline (Tet), kanamycin (Kan), and chloramphenicol (CMP). For this purpose, four sets of samples were prepared in triplicates and incubated at 37 °C. The first set was taken as a control in which only bacterial cultures were present to assess the growth kinetics in the absence of any treatments. In the second set, heavy metals were also added to the cultures. Antibiotics were supplemented to the bacterial cultures in the third set and the fourth set was comprised of heavy metals, antibiotics, and cultures. Before beginning the experiment, a set of rules was fixed; for example, the concentration of antibiotics in each experiment was fixed to 1 µl = 1 mL of broth. Three antibiotics, tetracycline (12.5 µg/mL), chloramphenicol (34 µg/mL), and kanamycin (50 µg/mL) were incorporated in the media. Concentrations of heavy metals were also fixed as 2 mM arsenic, 0.5 mM copper, and 0.5 mM lead. Using low concentration of heavy metals and standard concentration of antibiotics was already established in previous studies (Chen et al., 2015; Zhou et al., 2015). Finally, the optical density (OD₆₀₀) of the samples was measured through a UV/VIS spectrophotometer to study and compare the growth kinetics of the samples.

Molecular characterization

For molecular characterization, the 16S rRNA genes were amplified using the universal primers, 27F: AGA GTT TGA TCM TGG CTC AG, 515F: GTG CCA GCM GCC GCG GTA A and 1492R: CGG TTA CCT TGT TAC GAC TT. Colony PCR was performed in a 25-µL reaction using Taq buffer, 25 mM MgCl₂, 25 mM dNTPs, Taq polymerase 25 pmol/µL, primers, and 2 µL of DNA template. The PCR products were analyzed on 1% agarose gel and sent for commercial sequencing (MCLAB, USA) followed by analysis with GENTle version 1.9.4 (Manske, 2006). Bacterial isolates were identified by using BLAST (Johnson et al., 2008) and SILVA (Quast et al., 2013) databases based on the *e*-value and identity percentage. Phylogenetic analysis of strains was performed

by using BLAST (Johnson et al., 2008) in which the phylogenetic tree was generated by neighbor joining method. 16S rRNA gene sequences were submitted to GENBANK and the accession numbers were obtained.

Statistical analysis

Graph pad Prism 6 (version 6.00 for Windows, GraphPad Software, San Diego, California, USA, www.graphpad.com) was used for statistical analysis. Data were analyzed by using one way analysis of variance (ANOVA) to observe the effects of treatments on the growth kinetics of bacteria.

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Author contribution FK, MBK, and MBY designed experiments. MBK and MBY collected samples and conducted experiments under the supervision of FK; MBK drafted the manuscripts; FK revised the manuscript.

Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Conflict of interest The authors declare no competing interests.

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